

EXPERIMENT-12

Q1.Find the sum and product of first and last digits.

1. Problem Statement:

Write a menu-driven C program that accepts a positive integer and performs one of these operations:

Find the sum and product of the first and last digits.

Count the digits that are less than, equal to, and greater than a given digit k.

2. Algorithm:

Main Algorithm

Start.

Input a positive integer n.

Display the menu:

1. Find sum and product of first and last digits
2. Count digits lesser than, equal to, and greater than k

Input the user's choice.

Use a switch statement based on the choice.

If choice is 1:

Store n in a temporary variable temp.

Find the last digit using $\text{last} = n \% 10$.

Repeatedly divide temp by 10 until it becomes a single-digit number.

The remaining value of temp is the first digit.

Calculate $\text{sum} = \text{first} + \text{last}$.

Calculate $\text{product} = \text{first} * \text{last}$.

Display the first digit, last digit, sum, and product.

If choice is 2:

Input digit k.

Initialize $\text{lesser} = 0$, $\text{equal} = 0$, and $\text{greater} = 0$.

Store n in temp.

Repeat while $\text{temp} > 0$:

Find the last digit using $\text{digit} = \text{temp} \% 10$.

If $\text{digit} < k$, increase lesser by 1.

Else if digit == k, increase equal by 1.
 Else, increase greater by 1.
 Remove the last digit using temp = temp / 10.
 Display the three counts.
 For any other choice, display "Invalid choice".
 Stop.

3. Test Case Table

Test Case

Choice

Number

k

Expected Output

Actual Output

Result

1

1

12345

—

First=1, Last=5, Sum=6, Product=5

First=1, Last=5, Sum=6, Product=5

Pass

2

1

9876

—

First=9, Last=6, Sum=15, Product=54

First=9, Last=6, Sum=15, Product=54

Pass

3

1

505

—

First=5, Last=5, Sum=10, Product=25

First=5, Last=5, Sum=10, Product=25

Pass

4

2

12345

3

Lesser=2, Equal=1, Greater=2

Lesser=2, Equal=1, Greater=2

Pass

5

2

77727

7

Lesser=1, Equal=4, Greater=0

Lesser=1, Equal=4, Greater=0

Pass

6

2

24689

5

Lesser=2, Equal=0, Greater=3

Lesser=2, Equal=0, Greater=3

Pass

4. Program, Compilation and execution:



```
GNU nano 9.0
#include <stdio.h>

int main()
{
    int n, temp, choice;
    int first, last, sum, product;
    int k, digit;
    int lesser = 0, equal = 0, greater = 0;

    printf("Enter a positive integer: ");
    scanf("%d", &n);

    printf("\nMENU\n");
    printf("1. Find sum and product of first and last digits\n");
    printf("2. Count digits lesser than, equal to, and greater than k\n");

    printf("Enter your choice: ");
    scanf("%d", &choice);

    switch(choice)
    {
        case 1:
            last = n % 10;
            temp = n;

            while(temp >= 10)
            {
                temp = temp / 10;
            }

            first = temp;

            sum = first + last;
            product = first * last;

            printf("\nFirst digit = %d\n", first);
            printf("Last digit = %d\n", last);
            printf("Sum = %d\n", sum);
            printf("Product = %d\n", product);
        case 2:
            printf("Enter a value of k: ");
            scanf("%d", &k);

            while(n > 0)
            {
                digit = n % 10;

                if(digit < k)
                    lesser++;
                else if(digit == k)
                    equal++;
                else if(digit > k)
                    greater++;

                n = n / 10;
            }

            printf("\nLesser = %d, Equal = %d, Greater = %d\n", lesser, equal, greater);
    }
}
```

case 2:

```
printf("Enter digit k: ");
scanf("%d", &k);
```

```
temp = n;
```

```
while(temp > 0)
```

```
{
```

```
    digit = temp % 10;
```

```
    if(digit < k)
```

```
    {
```

```
        lesser++;
```

```
    }
```

```
    else if(digit == k)
```

```
    {
```

```
        equal++;
```

```
    }
```

```
    else
```

```
    {
```

```
        greater++;
```

```
    }
```

```
    temp = temp / 10;
```

```
}
```

```
printf("\nCount_lesser = %d\n", lesser);
```

```
printf("Count_equal = %d\n", equal);
```

```
printf("Count_greater = %d\n", greater);
```

```
break;
```

```
default:
```

```
    printf("\nInvalid choice\n");
```

```
}
```

```
return 0;
```

```
}
```

```
Session Actions Edit View Help
(kali@kali)-[~]
$ nano 12a.c
(kali@kali)-[~]
$ gcc 12a.c -o 12a
(kali@kali)-[~]
$ ./12a
Enter a positive integer: 12345

MENU
1. Find sum and product of first and last digits
2. Count digits lesser than, equal to, and greater than k
Enter your choice: 1

First digit = 1
Last digit = 5
Sum = 6
Product = 5

(kali@kali)-[~]
$ ./12a
Enter a positive integer: 12345

MENU
1. Find sum and product of first and last digits
2. Count digits lesser than, equal to, and greater than k
Enter your choice: 2
Enter digit k: 3

Count_lesser = 2
Count_equal = 1
Count_greater = 2

(kali@kali)-[~]
$
```

6. Conclusion

The program was compiled and tested using the prepared test cases. It correctly uses a switch-case menu, extracts the first and last digits for Choice 1, and compares each digit with k for Choice 2. All the prepared test cases produced the expected results.

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